Zackory Erickson

zackory@cmu.edu
zackory.com

Carnegie Mellon University
Robotics Institute
5000 Forbes Avenue
Pittsburgh, PA 15213



Figure : Real-world deployments of research from our lab with populations that can benefit from robotic assistance, including individuals with spinal cord injury, stroke, cerebral palsy, amputation, and muscle dystrophy or frailty associated with aging. Images span robotic caregiving and physical human-robot interaction in applications including eating and drinking, hygiene and skin care, dressing, bedding manipulation, lift and transfer, and door opening with a robotic wheelchair.

Current Positions

Carnegie Mellon University, Robotics Institute
Raj Reddy Assistant Professor — 2026–present
Assistant Professor — 2021–2026
Courtesy appointment: Biomedical Engineering

Founding Scientific Advisor, Intuitive Autonomy — 2025–present

Advisor, Skylark Labs — 2026–present

Education

Ph.D., Robotics — 2016–2021
Georgia Institute of Technology
Advisor: Charles C. Kemp

M.S., Computer Science — 2020
Georgia Institute of Technology
Advisor: Charles C. Kemp

B.S., Computer Science, Mathematics (double major) — 2012–2016
University of Wisconsin–La Crosse

Honors and Awards

Raj Reddy Career Development Professorship in Robotics 2026

NSF CAREER 2026
"CAREER: Scaling Up Physical Human-Robot Interaction with Generative Simulation"

Best Paper Award, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026
"WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding"

Best Paper Finalist, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026
"Bidirectional Human-Robot Communication for Physical Human-Robot Interaction"

Best Paper Finalist, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2025
"LAMS: LLM-Driven Automatic Mode Switching for Assistive Teleoperation"

OSCAR Scholar, University of Pittsburgh, Alzheimer's Disease Research Center (ADRC) 2024

Best Paper Award, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2024
"Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator"

Best Poster Award, CMU Mechanical Engineering Research Symposium 2024
"Extended Kalman Filter for Real-Time State Estimation of Cloth for Robot Manipulation"

NVIDIA Academic Hardware Grant 2022
"Adapting to Distribution Shift in Deformable Manipulation with Assistive Robots"

NVIDIA Fellowship Finalist 2020

Best Student Paper Award, IEEE International Conference on Rehabilitation Robotics (ICORR) 2019
"Multidimensional Capacitive Sensing for Robot-Assisted Dressing and Bathing"

Best Paper Award in Service Robotics Finalist, IEEE ICRA 2019
"Classification of Household Materials via Spectroscopy"

President's Fellowship, Georgia Tech 2016–2020

4th Heidelberg Laureate Forum 2016

Honorable Mention, NSF GRFP 2016

Strzelczyk Award 2016
Awarded to the top graduating senior in the College of Science and Health for academic achievement and service to the campus and community.

MIT CONVERGE 2015
One of 18 prospective PhD students in the nation invited to tour MIT.

Berkeley Engineering Preview Days 2015
One of 14 prospective PhD students nationwide invited to tour UC Berkeley.

Grace Olwell Memorial Endowment Fund Scholarship 2015

Xcel Energy Scholarship 2015

John and Lois Storlie Scholarship in Computer Science 2014

Undergraduate Research Grant, UW–La Crosse 2013

Scottish Rite Abbott Scholarship 2013

Dean's List, UW–La Crosse 8 semesters

Mentoring

PhD Students

Cailyn Smith, CMU, RI 2025–

Yucheng Li, CMU, Heinz 2025–

Junxiang (Jim) Wang, CMU, RI 2024–

Zheyuan Hu, CMU, RI (co-advised with Aviral Kumar) 2024–

Hongyi Chen, CMU, RI (co-advised with Jeffrey Ichnowski) 2024–

Kavya Puthuveetil, CMU, RI (co-advised with Reid Simmons) 2022–

Zulekha Karachiwalla, CMU, RI (co-advised with Henny Admoni) 2022–2025

Je-Han Yang, CMU, BME (co-advised with Douglas Weber) 2022–

Yufei Wang, CMU, RI (co-advised with David Held) 2021–

Akhil Padmanabha, CMU, RI (co-advised with Carmel Majidi) 2021–2025
Current: Meta

M.S. Students

Niharika Pant, CMU, MechE 2025–

Yitong (Elly) Wang, CMU, MLD 2025–

Kshitiz Kshitiz, CMU, RI 2025–

Saksham Bhutani, CMU, RI 2025–

Shikun Ban, CMU, RI 2025–

Shohei Nagai, CMU, RI 2025–

Tianhui (Tiffany) Han, CMU, RI 2025–

Anjana Ramachandran, CMU, BME-AIE 2025–

Julian Millan, CMU, MechE 2025–

Mengning Wu, CMU, RI 2025–

Qihao Lin, CMU, BME 2025–

Yinong He, CMU, RI 2025–

Crystal Zhou, CMU, MechE 2025–

MRSD Project: Operating room logistics 2.0, CMU, RI 2025–
Students: Ayush Sharan, Saketh Polamraju, Akshit Raizada, Mehar Goenka, Teja Kanthamneni

Hanqi Feng, CMU, RI 2024–

Alexis Hao, CMU, RI 2024–

Tiancheng (Tony) Wu, CMU, RI 2024–

Mino Nakura, CMU, RI 2024–

Pratik Bhowal, CMU, RI 2024–

Divyam Goel, CMU, RI 2024–

Xinwen Xu, CMU, MechE 2024–

Aditi Guruprasad, University of Pittsburgh, Bioengineering 2024–2025

Jing Gao, CMU, RI 2024–2025

Zoe LaLena, CMU, RI 2024–2025

Yatharth Ahuja, CMU, RI 2023–2024

Yiran Tao, CMU, RI 2023–2025
Current: PhD Student at MIT

Michaela Tecson, CMU, RI 2023–

Tariq Hussain, CMU, MechE 2023–2025

Yexin Hu, CMU, CEE 2023–2025
Current: PhD Student at TU Wien

Abhishek Tandon, CMU, RI 2023–2024
Current: Adobe

Anujraaj (Argo) Goyal, CMU, RI 2023–2024
Current: Snapchat

Zhanyi Sun, CMU, RI 2023–2024
Current: PhD Student at Stanford

MRSD Project: Operating room logistics, CMU, RI 2023–2025
Students: Gaurav Sethia, Jinkai Qiu, Roman Kaufman, Tanmay Agarwal, Yungjun Kim, Siddharth Ghodasara

Sankalp Chopkar, CMU, RI 2022–2024

Atharva Kusalkar, CMU, RI 2022–2024

Saurav Kambil, CMU, MechE 2022–2024

Zilin Zhang, CMU, RI 2022–2024

MRSD Project: Auxilio, CMU, RI 2022–2023
Students: Shaolin Kataria, Shivam Tripathy, Praveen Venkatesh, Abhinav Gupta, Atharva Pusalkar

Fukang Liu, CMU, MechE 2021–2023
Current: PhD Student at Georgia Tech

Vaidehi Patil, CMU, RI 2021–2022

MRSD Project: TouRI, CMU, RI 2021–2022
Students: Shivani Sivakumar, Jashkumar Diyora, Shruti Gangopadhyay, Prakhar Pradeep, Jigarkumar Patel

Pratyusha Karnati, Georgia Tech, CS 2020–2021
Current: Google X Robotics

Yijun (Esther) Gu, Georgia Tech, CS 2019–2021
Current: PhD Student at Imperial College London

Undergraduate Students

Nir Pechuk, CMU, CS 2026–

Naveen Enock, CMU, CIT 2025–

Jasmine Li, CMU, ECE 2025–

Robyn Wu, CMU, CS 2025–

Riya Kadakia, CMU, CS 2025

Karolena Zhou, CMU, CIT 2024–

Divya Gupta, CMU, CS 2024–

Lehong Wu, Peking University, CS 2024–

Vivianna Lieu, CMU, CS 2024–

Annika Srinivasan, UIUC, MechE 2024–2025

Cindy Sun, CMU, MechE 2024–

Victoria de Leon, ITESM (NSF REU, RISS), Robotics 2024–2025

Tanisha Mehta, CMU, ECE, CS 2024–2025

Catherine Li, CMU, CS 2024–

Navin Sriram, Indian Insitute of Technology Madras, CS 2024–2025

Sreyas Venkataraman, Indian Institute of Technology, Kharagpur, CS 2024–2025

Ziyu Wang, Tsinghua University, IIIS 2024–2025

Bharath Hegde, BITS Pilani University, CS 2024–2025

Zhenghao (Cary) Jin, CMU, ECE 2024–2025

Glenda Tan, CMU, CS 2024–

Jessica Han, CMU, CS 2024–

Ryan Ding, CMU, CS 2024–

Pranavi Kondapalli, CMU, CS 2024–

Maxwell Soh, CMU, CS 2023–2024

Ellen Lee, CMU, CS 2023–2024

Weikang Wan, Peking University, CS 2023–2024

Chen Chen, Tsinghua University, CS 2023–2024

Janavi Gupta, CMU, CS 2023–

Sumayya Syeda, CMU, CIT 2023–2024

Preethi Krishnamoorthy, CMU, CS 2023–2024

Jessie Yuan, CMU, CS 2023–2024

Madeliene Brutin, CMU, CS 2023–2024

Kendra Givens, Middle Tennessee State University (NSF REU, RISS), CS 2023–2024
Current: PhD student at Vanderbilt University

Alexandra Gillespie, Colby College (NSF REU, RISS), CS 2023–2024
Current: PhD student at University of Edinburgh

Yikang (Bruce) Cheng, CMU, CS 2022–2023

Jacob Delgado-López, University of Puerto Rico (NSF REU, RISS), CS 2022
Current: PhD student at JHU

Wesley Lewis, University of Virginia (NSF REU, RISS), CS 2022–2024

Allen Zheng, CMU, CS 2022–2024

Qin (Alicia) Wang, CMU, CS 2022–2023

Daphne Han, CMU, CB 2022

Alexandra (Sasha) Wald, CMU, CS 2021–2024

Kavya Puthuveetil, Virginia Commonwealth University (NSF REU), BME 2021–2022

Samantha Mutiti, Georgia Tech, BME 2021

Holden Schaffer, Georgia Tech, CS 2020–2021

Siyan (Sylvia) Li, Georgia Tech, CS 2018–2019

Jiaqi (Julia) Chen, Georgia Tech, CS 2018
Current: PhD Student at ETH Zurich

Katelyn Sosnowski, University of Arizona (NSF REU), BME 2018
Current: BME PhD Student at University of Arizona

Mallak Taleb, University of Michigan (NSF REU), BME 2018

Bharat Srirangam, Georgia Tech, CS 2018–2020
Current: Woot, Inc.

Eliot Xing, Georgia Tech, CE 2017–2022

Vamsee Gangaram, Georgia Tech, CS 2017–2020
Current: Microsoft

Jong Hwa (Austin) Jang, Georgia Tech, CS 2017–2018

Maggie Collier, University of Alabama at Birmingham (NSF REU), BME 2017
Current: Robotics PhD student at CMU, NDSEG fellow

Nathan Luskey, Georgia Tech, BME 2017–2018
Current: MSCS student at CMU

Teaching

16-467: Introduction to Human-Robot Interaction, CMU Spring 2025

16-762: Mobile Manipulation, CMU Spring 2024, 2026–

16-741: Mechanics of Manipulation, CMU Fall 2022–

16-887: Robotic Caregivers and Intelligent Physical Collaboration, CMU Spring 2022–2023

Robotic Caregivers (BMED 4833/8813), Co-instructor, Georgia Tech Spring 2021

Robotic Caregivers (BMED 4803/8813), Co-developer and instructor, Georgia Tech Spring 2020

Invited Talks

Robotic Caregivers and Wearable Health Sensors
Samsung Research America 2025
Allegheny Health Network 2025
Shanghai Jiao Tong University School of Medicine 2025

Physical HRI with Deformable Manipulation

ICRA 2025 Workshop on Reflections on Representations and Manipulating Deformable Objects 2025

Robot Learning and Wearable Interfaces in Pursuit of Robotic Caregivers
University of Wisconsin–Madison 2025
Nankai University 2025
National University of Singapore 2025
University of Texas at Austin 2024

Robot Learning, Wearable Sensing, and Teleoperation in Pursuit of Robotic Caregivers
16-761: Mobile Robotics, CMU 2024
CMU RI Seminar 2023
RISS RoboLaunch 2023
Northeastern University 2023
University of Utah 2023
University of Illinois Urbana-Champaign 2023

My Journey to Professor
Robotics Institute Summer Scholars Seminar, CMU 2023

Robotics and Artificial Intelligence in Healthcare and Opportunities for Innovation
Clinical Impact Symposia, BrightSpring Health Services 2023

Capacitive Servoing and Spectroscopy for Physically Assistive Robotics
16-722: Sensing and Sensors, CMU 2022

Haptic Perspective-taking from Vision and Force
CMU RI Seminar 2022

Capacitive Proximity Servoing for Physically Assistive Robotics
RSS 2022 Workshop on Advances in Close Proximity Human-Robot Collaboration 2022

Robotic Caregiving and Human Interaction
24-675: Humanoid Robotics and Cognition, CMU 2022
16-311: Introduction to Robotics, CMU 2022
05-899: Special Topics in HCI: Accessibility, CMU 2021

Capacitive Servoing for Physically Assistive Robotics
IROS 2021 4th Workshop on Proximity Perception in Robotics 2021

Robotic Caregivers—Sensing, Simulation, and Physical Human-Robot Interaction
Carnegie Mellon University 2021
University of Pennsylvania 2021

Physics-based Cloth Simulation and Learning Towards Robotic Caregiving
ICRA 2021 Workshop on Representing and Manipulating Deformable Objects 2021

Robotic Caregivers—Recent Advances in Physics-based Simulation
Medical Robotics Club, Georgia Tech 2021

Robot-Assisted Dressing
RESNA 2018 Workshop on Smart and Robotic Homes 2018

Multimodal Anomaly Detection
Mathematics Colloquium, UW–La Crosse 2015

Academic Service

Chair of PhD Admissions Committee — CMU, Robotics Institute 2025–

Workshop Organizer — IROS, RObotic MAnipulation of Deformable Objects (ROMADO) 2025

Area Chair — Robotics: Science and Systems (RSS) 2025

Workshop Organizer — HRI, Enhancing Independence with Physical Caregiving Robots 2025

Co-Chair of PhD Admissions Committee — CMU, Robotics Institute 2024

Workshop Organizer — RSS, Learning for Assistive Robotics 2024

Faculty Senate — CMU, representing the Robotics Institute 2024–

Advisory board — University of Wisconsin–La Crosse, Computer Science Department 2024–

Panelist — HRI, Workshop on HRI for Aging in Place 2024

Workshop Organizer — ICRA, Exploring Role Allocation in Human-Robot Co-Manipulation 2024

Associate Editor — IEEE Robotics and Automation Letters (RA-L) 2023–
Manipulation and Grasping

NSF AI-CARING Arena Lead — NSF AI Institute 2023–2024
Arena on AI for Interactive Coaches and Services

Virtual Experience Chair — Conference on Robot Learning (CoRL) 2023

Seminar and Panel Organizer, Life as a Professor, CMU 2023
Semesterly seminar where a panel of CMU faculty discuss their perspectives on becoming and being a professor for graduate students interested in the academic career path

Special Session Organizer — RO-MAN, Special Session on Human-Agent/Robot Interaction in Healthcare and Medicine 2023

Workshop Organizer — ICRA, Emerging paradigms for assistive robotic manipulation: from research labs to the real world 2023

PhD Admissions Committee — CMU, Robotics Institute 2022–

Chair — IROS 2022 session on Art and Entertainment and Manipulation 2022

Co-Chair — ICRA 2022 session on Physical HRI 2022

Area Chair — Conference on Robot Learning (CoRL) 2022–2023

Workshop Organizer — IROS, 5th Workshop on Proximity Perception - Towards Next-Generation Multi-Modal Sensing in Soft Structures 2022

Associate Editor
IEEE Conference on Robot and Human Interactive Communication (RO-MAN) 2021, 2022

Reviewer
T-RO, ICRA, IROS, RA-L, CoRL, CVPR, ICML, ICLR, NeurIPS, Humanoids, HRI, Sensors, RO-MAN, Science Robotics, RSS Pioneers

Workshop Organizer — ICRA, Learning for Caregiving Robots 2021

Seminar Organizer, Georgia Tech
Life as a Professor: Student Advising and Recruiting 2020
Life as a Professor: Funding 101 2019
Life as a Professor: Starting a Start-Up 2018

Panelist
Graduate Intro to Robotics Research, Georgia Tech 2018

Summer Undergraduate Research Experience (SURE) Program, Georgia Tech 2017–2019

RoboGrads (robotics graduate student organization), Georgia Tech
Vice President for Robotics PhD 2019–2020
President 2018–2019

Student Thesis Committees

Vihaan Misra, CMU, MSR 2025
Caleb Escobedo, CU Boulder, PhD 2024–
Christian Berger, CMU, MSR 2024
Christopher Chang, CMU, MSR 2024
Zhiyang (Jerry) He, UC Berkeley, PhD 2023–

Student Qualifying Exam Committees

Maggie Collier, CMU, PhD 2024
Ruixuan Liu, CMU, PhD 2023

Outreach

CEO Soak, ALS Association, Pittsburgh 2025
Organized a table showcasing robotics research at CMU that can impact individuals with ALS.

AI Scholars, Co-organizer, CMU 2023–
Each summer, AIS provides 30+ rising high school seniors who have historically been excluded from STEM fields the opportunity to explore artificial intelligence through 4 weeks of courses and hands-on projects with Carnegie Mellon faculty, staff, and researchers.

AI4ALL, Co-organizer, CMU 2022

RoboGrads, Vice President of Outreach, Georgia Tech 2017–2018
Organized K-12 outreach events for over 30 robotics labs at Georgia Tech

Biomedical Robotics Club, Mentor, Georgia Tech 2016–2019
Mentoring over 50 undergraduate students in how to research and build assistive devices and robots for people with impairments.

FIRST Lego League, Judge 2015–2021

CS Outreach & Diversity Club, UW–La Crosse 2015–2016
Organizing CS and robotics events for K-12 students.

FIRST Robotics, Mentor, Central High School and Holmen High School 2012–2016

Peer-Reviewed Publications (Conferences and Journals)

[1] **Bidirectional Human-Robot Communication for Physical Human-Robot Interaction**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2026 **(Best Paper Finalist)**
Junxiang Wang, Cindy Wang, Rana Soltani Zarrin, Zackory Erickson

[2] **WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2026 **(Best Paper Award)**
Akhil Padmanabha*, Jessie Yuan*, Tanisha Mehta, Rajat Kumar Jenamani, Eric Hu, Victoria de León, Anthony Wertz, Janavi Gupta, Ben Dodson, Yunting Yan, Carmel Majidi, Tapomayukh Bhattacharjee†, Zackory Erickson†

[3] **Knowing When to Help: A Framework for Error Detection in Meal Preparation Tasks for Older Adults**
ACM/IEEE International Conference on Human-Robot Interaction (HRI) Late Breaking Reports, 2026
Michaela Tecson, Zackory Erickson, and Reid Simmons

[4] **AI in geriatric psychiatry: precision meets human experience**
Neuropsychopharmacology, 2026
Akiko Mizuno, Zackory Erickson, Daniel E. Jimenez, and Howard Jay Aizenstein

[5] **In Vivo Skin 3-D Surface Reconstruction and Wrinkle Depth Estimation Using Handheld High Resolution Tactile Sensing**
Advanced Healthcare Materials, 2025
Akhil Padmanabha, Arpit Agarwal, Catherine Li, Austin Williams, Dinesh K Patel, Sankalp Chopkar, Achu Wilson, Ahmet Ozkan, Wenzhen Yuan, Sonal Choudhary, Arash Mostaghimi, Zackory Erickson, Carmel Majidi

[6] **Geometric Red-Teaming for Robotic Manipulation**
Conference on Robot Learning (CoRL), 2025 **(Oral Presentation)**
Divyam Goel, Yufei Wang, Tiancheng Wu, Helen Qiao, Pavel Piliptchak, David Held†, Zackory Erickson†

[7] **Force-Modulated Visual Policy for Robot-Assisted Dressing with Arm Motions**
Conference on Robot Learning (CoRL), 2025
Alexis Yihong Hao, Yufei Wang, Navin Sriram Ravie, Bharath Hegde, David Held†, Zackory Erickson†

[8] **CoRI: Synthesizing Communication of Robot Intent for Physical Human-Robot Interaction**
Conference on Robot Learning (CoRL), 2025
Junxiang Wang, Emek Barış Küçüktabak, Rana Soltani Zarrin, Zackory Erickson

[9] **ORB: Operating Room Bot, Automating Operating Room Logistics through Mobile Manipulation**
International Conference on Automation Science and Engineering (CASE), 2025
Jinkai Qiu*, Yungjun Kim*, Gaurav Sethia*, Tanmay Agarwal*, Siddharth Ghodasara*, Zackory Erickson, Jeffrey Ichnowski

[10] **Leveraging Large Language Models for Preference-Based Sequence Prediction**
International Conference on Agents and Artificial Intelligence (ICAART 2025), 2025
Michaela Tecson, Daphne Chen, Michelle Zhao, Zackory Erickson, Reid Simmons

[11] **Incremental Learning for Robot Shared Autonomy**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Yiran Tao, Guixiu Qiao, Dan Ding, Zackory Erickson

[12] **Real-World Offline Reinforcement Learning from Vision Language Model Feedback**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Sreyas Venkataraman*, Yufei Wang*, Ziyu Wang, Navin Sriram Ravie, Zackory Erickson†, David Held†

[13] **SkinGrip: An Adaptive Soft Robotic Manipulator with Capacitive Sensing for Whole-Limb Bed Bathing Assistance**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Fukang Liu, Kavya Puthuveetil, Akhil Padmanabha, Karan Khokar, Zeynep Temel, Zackory Erickson

[14] **DiSRT-In-Bed: Diffusion-Based Sim-to-Real Transfer Framework for In-Bed Human Mesh Recovery**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
Jing Gao, Ce Zheng, Laszlo A Jeni, Zackory Erickson

[15] **RoboCAP: Robotic Classification and Precision Pouring of Diverse Liquids and Granular Media with Capacitive Sensing**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025

Yexin Hu*, Alexandra Gillespie*, Akhil Padmanabha, Kavya Puthuveetil, Wesley Lewis, Karan Khokar, Zackory Erickson

[16] **ArticuBot: Learning Universal Articulated Object Manipulation Policy via Large Scale Simulation**
Robotics Science and Systems (RSS), 2025
Yufei Wang*, Ziyu Wang*, Mino Nakura, Pratik Bhowal, Chia-Liang Kuo, Yi-Ting Chen, Zackory Erickson†, David Held†

[17] **High-density Electromyography for Effective Gesture-based Control of Physically Assistive Mobile Manipulators**
npj Robotics, 2025
Jehan Yang, Kent Shibata, Douglas Weber, Zackory Erickson

[18] **LAMS: LLM-Driven Automatic Mode Switching for Assistive Teleoperation**
ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025 **(Best Paper finalist)**
Yiran Tao*, Jehan Yang*, Dan Ding, Zackory Erickson

[19] **EMGBench: Benchmarking Out-of-Distribution Generalization and Adaptation for Electromyography**
Advances in neural information processing systems (NeurIPS), 2024
Jehan Yang, Maxwell Soh, Vivianna Lieu, Douglas J Weber, Zackory Erickson

[20] **VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots**
ACM Symposium on User Interface Software and Technology (UIST), 2024
Akhil Padmanabha*, Jessie Yuan*, Janavi Gupta, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, Zackory Erickson

[21] **DiffTORI: Differentiable Trajectory Optimization for Deep Reinforcement and Imitation Learning**
Advances in neural information processing systems (NeurIPS), 2024 **(Spotlight Presentation)**
Weikang Wan*, Ziyu Wang*, Yufei Wang*, Zackory Erickson†, David Held†

[22] **RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation**
International Conference on Mahince Learning (ICML), 2024
Yufei Wang*, Zhou Xian*, Feng Chen*, Tsun-Hsuan Wang, Yian Wang, Katerina Fragkiadaki, Zackory Erickson, David Held, Chuang Gan

[23] **RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback**
International Conference on Mahince Learning (ICML), 2024
Yufei Wang*, Zhanyi Sun*, Jesse Zhang, Zhou Xian, Erdem Bıyık, David Held†, Zackory Erickson†

[24] **Do Mistakes Matter? Comparing Trust Responses of Different Age Groups to Errors Made by Physically Assistive Robots**
IEEE International Conference on Robot and Human Interactive Communication (ROMAN), 2024
Sasha Wald*, Kavya Puthuveetil*, Zackory Erickson

[25] **Unfolding the literature: A review of robotic cloth manipulation**
Annual Review of Control, Robotics, and Autonomous Systems, 2024
Alberta Longhini, Yufei Wang, Irene Garcia-Camacho, David Blanco-Mulero, Marco Moletta, Michael Welle, Guillem Alenyà, Hang Yin, Zackory Erickson, David Held, Júlia Borràs, Danica Kragic

[26] **AdaFold: Adapting Folding Trajectories of Cloths via Feedback-Loop Manipulation**
IEEE Robotics and Automation Letters (RA-L), 2024
Alberta Longhini, Michael C. Welle, Zackory Erickson, Danica Kragic

[27] **BodyMAP-Jointly Predicting Body Mesh and 3D Applied Pressure Map for People in Bed**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
Abhishek Tandon*, Anujraaj Goyal*, Henry M Clever, Zackory Erickson

[28] **Force Constrained Visual Policy: Safe Robot-Assisted Dressing via Multi-Modal Sensing**
IEEE Robotics and Automation Letter (RA-L), 2024
Zhanyi Sun*, Yufei Wang*, David Held†, Zackory Erickson†

[29] **Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2024 **(Best Paper Award)**
Akhil Padmanabha, Janavi Gupta, Chen Chen, Jehan Yang, Vy Nguyen, Douglas J Weber, Carmel Majidi, Zackory Erickson

[30] **Quantifying Assistive Robustness Via the Natural-Adversarial Frontier**
Conference on Robot Learning (CoRL), 2023
Jerry Zhi-Yang He, Daniel S Brown, Zackory Erickson, Anca Dragan

[31] **A Multimodal Sensing Ring for Quantification of Scratch Intensity**
Nature Communications Medicine, 2023
Akhil Padmanabha, Sonal Choudhary, Carmel Majidi, Zackory Erickson

[32] **Robust Body Exposure (RoBE): A Graph-based Dynamics Modeling Approach to Manipulating Blankets over People**
IEEE Robotics and Automation Letters (RA-L), 2023
Kavya Puthuveetil, Sasha Wald, Atharva Pusalkar, Pratyusha Karnati, and Zackory Erickson

[33] **One Policy to Dress Them All: Learning to Dress People with Diverse Poses and Garments**
Robotics Science and Systems (RSS), 2023
Yufei Wang, Zhanyi Sun, Zackory Erickson†, David Held†

[34] **HAT: Head-Worn Assistive Teleoperation of Mobile Manipulators**
IEEE International Conference on Robotics and Automation (ICRA), 2023
Akhil Padmanabha*, Qin Wang*, Daphne Han, Jashkumar Diyora, Kriti Kacker, Hamza Khalid, Liang-Jung Chen, Carmel Majidi, Zackory Erickson

[35] **SLURP! Spectroscopy of Liquids Using Robot Pre-Touch Sensing**
IEEE International Conference on Robotics and Automation (ICRA), 2023
Nathaniel Hanson*, Wesley Lewis*, Kavya Puthuveetil, Donelle Furline Jr, Akhil Padmanabha, Taskin Padir, Zackory Erickson

[36] **EDO-Net: Learning Elastic Properties of Deformable Objects from Graph Dynamics**
IEEE International Conference on Robotics and Automation (ICRA), 2023
Alberta Longhini*, Marco Moletta*, Alfredo Reichlin, Michael C. Welle, David Held, Zackory Erickson, Danica Kragic

[37] **Elastic Context: Encoding Elasticity for Data-driven Models of Textiles**
IEEE International Conference on Robotics and Automation (ICRA), 2023
Alberta Longhini, Marco Moletta, Alfredo Reichlin, Michael C. Welle, Alexander Kravberg, Yufei Wang, David Held, Zackory Erickson, Danica Kragic

[38] **A Study of Causal Confusion in Preference-Based Reward Learning**
International Conference on Learning Representations (ICLR), 2023
Jeremy Tien, Jerry Zhi-Yang He, Zackory Erickson, Anca D. Dragan, Daniel S. Brown

[39] **ToolFlowNet: Robotic Manipulation with Tools via Predicting Tool Flow from Point Clouds**
Conference on Robot Learning (CoRL), 2022
Daniel Seita, Yufei Wang, Edward Yao Li, Sarthak J Shetty, Zackory Erickson, and David Held

[40] **Learning Representations that Enable Generalization in Assistive Tasks**
Conference on Robot Learning (CoRL), 2022
Jerry Zhi-Yang He, Zackory Erickson, Daniel S. Brown, Aditi Raghunathan, and Anca Dragan

[41] **Characterization of a Meso-Scale Wearable Robot for Bathing Assistance**
IEEE International Conference on Robotics and Biomimetics (ROBIO), 2022
Fukang Liu, Vaidehi Patil, Zackory Erickson, and Zeynep Temel

[42] **Visual Haptic Reasoning: Estimating Contact Forces by Observing Deformable Object Interactions**
IEEE Robotics and Automation Letters (RA-L), 2022
Yufei Wang, David Held, and Zackory Erickson

[43] **CapSense: A Real-Time Capacitive Sensor Simulation Framework for Physical Human-Robot Interaction**
IEEE Robotics and Automation Letters (RA-L), 2022
Christian Schöffmann, Zackory Erickson, and Hubert Zangl

[44] **Bodies Uncovered: Learning to Manipulate Real Blankets Around People via Physics Simulations**
IEEE Robotics and Automation Letters (RA-L), 2022
Kavya Puthuveetil, Charles C. Kemp, and Zackory Erickson

[45] **Characterizing Multidimensional Capacitive Servoing for Physical Human-Robot Interaction**
IEEE Transactions on Robotics (T-RO), 2022
Zackory Erickson, Henry M. Clever, Vamsee Gangaram, Eliot Xing, Greg Turk, C. Karen Liu, and Charles C. Kemp

[46] **Assistive VR Gym: Interactions with Real People to Improve Virtual Assistive Robots**
IEEE Conference on Robot and Human Interactive Communication (RO-MAN), 2020
Zackory Erickson*, Yijun Gu*, and Charles C. Kemp

[47] **Multimodal Material Classification for Robots using Spectroscopy and High Resolution Texture Imaging**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020
Zackory Erickson, Eliot Xing, Bharat Srirangam, Sonia Chernova, and Charles C. Kemp

[48] **Bodies at Rest: 3D Human Pose and Shape Estimation from a Pressure Image using Synthetic Data**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020 **(Oral)**
Henry M. Clever, Zackory Erickson, Ariel Kapusta, Greg Turk, C. Karen Liu, and Charles C. Kemp

[49] **Assistive Gym: A Physics Simulation Framework for Assistive Robotics**
IEEE International Conference on Robotics and Automation (ICRA), 2020
Zackory Erickson, Vamsee Gangaram, Ariel Kapusta, C. Karen Liu, and Charles C. Kemp

[50] **Learning to Collaborate from Simulation for Robot-Assisted Dressing**
IEEE Robotics and Automation Letters (RA-L), 2020
Alexander Clegg, Zackory Erickson, Patrick Grady, Greg Turk, Charles C. Kemp, and C. Karen Liu

[51] **Active Robot-Assisted Feeding with a General-Purpose Mobile Manipulator: Design, Evaluation, and Lessons Learned**
Robotics and Autonomous Systems, 2020
Daehyung Park, Yuuna Hoshi, Harshal P. Mahajan, Ho Keun Kim, Zackory Erickson, Wendy A. Rogers, Charles C. Kemp

[52] **Multidimensional Capacitive Sensing for Robot-Assisted Dressing and Bathing**
IEEE International Conference on Rehabilitation Robotics (ICORR), 2019 **(Best Student Paper)**
Zackory Erickson, Henry M. Clever, Vamsee Gangaram, Greg Turk, C. Karen Liu, and Charles C. Kemp

[53] **Classification of Household Materials via Spectroscopy**
IEEE Robotics and Automation Letters (RA-L), 2019 **(Best Paper Award in Service Robotics**

Finalist at ICRA 2019)
Zackory Erickson, Nathan Luskey, Sonia Chernova, and Charles C. Kemp

[54] **Personalized Collaborative Plans for Robot-Assisted Dressing via Optimization and Simulation**
Autonomous Robots, 2019
Ariel Kapusta, Zackory Erickson, Henry M. Clever, Wenhao Yu, C. Karen Liu, Greg Turk, and Charles C. Kemp

[55] **Autonomous Tool Construction Using Part Shape and Attachment Prediction**
Robotics: Science and Systems (RSS), 2019
Lakshmi Nair, Nithin Srikanth, Zackory Erickson, Sonia Chernova

[56] **Deep Haptic Model Predictive Control for Robot-Assisted Dressing**
IEEE International Conference on Robotics and Automation (ICRA), 2018
Zackory Erickson, Henry M. Clever, Greg Turk, C. Karen Liu, and Charles C. Kemp

[57] **Tracking Human Pose During Robot-Assisted Dressing using Single-Axis Capacitive Proximity Sensing**
IEEE Robotics and Automation Letters (RA-L), 2018
Zackory Erickson, Maggie Collier, Ariel Kapusta, and Charles C. Kemp

[58] **3D Human Pose Estimation on a Configurable Bed from a Pressure Image**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2018
Henry M. Clever, Ariel Kapusta, Daehyung Park, Zackory Erickson, Yash Chitalia, Charles C. Kemp

[59] **Semi-Supervised Haptic Material Recognition for Robots using Generative Adversarial Networks**
Conference on Robot Learning (CoRL), 2017
Zackory Erickson, Sonia Chernova, and Charles C. Kemp

[60] **What Does the Person Feel? Learning to Infer Applied Forces During Robot-Assisted Dressing**
IEEE International Conference on Robotics and Automation (ICRA), 2017
Zackory Erickson, Alexander Clegg, Wenhao Yu, Greg Turk, C. Karen Liu, and Charles C. Kemp

[61] **Learning to Navigate Cloth using Haptics**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
Alexander Clegg, Wenhao Yu, Zackory Erickson, C. Karen Liu, and Greg Turk

[62] **A Multimodal Execution Monitor with Anomaly Classification for Robot-Assisted Feeding**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
Daehyung Park, Hokeun Kim, Yuuna Hoshi, Zackory Erickson, Ariel Kapusta, and Charles C. Kemp

[63] **Multimodal Execution Monitoring for Anomaly Detection During Robot Manipulation**
IEEE International Conference on Robotics and Automation (ICRA), 2016
Daehyung Park, Zackory Erickson, Tapomayukh Bhattacharjee, and Charles C. Kemp

Workshop Papers

[1] **BlenDAgger: Blended Shared Control for Interactive Imitation Learning**
IROS 2026: Workshop on Building Scalable Infrastructure for Robot Learning: From Data Scaling to Real-World Deployment, 2026
Cailyn Smith, Geoffrey Sun, Henny Admoni, and Zackory Erickson

[2] **Bimanual Wearable EMG Control for Multi-Day In-Home Mobile Manipulation by a User with Quadriplegia**
HRI 2026: Workshop on Lifelong Learning and Personalization in Long-Term Human-Robot Interaction,

2026
Jehan Yang, Eleanor Hodgson, Cindy Sun, Zackory Erickson and Douglas J. Weber

[3] **RaC: Robot Learning for Long-Horizon Tasks by Scaling Recovery and Correction**
CoRL 2025: Workshop on Making Sense of Data in Robotics: Composition, Curation, and Interpretability at Scale, 2025
Zheyuan Hu, Robyn Wu, Naveen Enock, Jasmine Jia-ni Li, Riya Kadakia, Zackory Erickson, Aviral Kumar

[4] **Towards Wearable Interfaces for Robotic Caregiving**
HRI 2025: Workshop on Physical Caregiving Robots, 2025
Akhil Padmanabha, Carmel Majidi, Zackory Erickson

[5] **Towards an LLM-Based Speech Interface for Robot-Assisted Feeding**
UIST 2024 demo track: Symposium on User Interface Software and Technology, 2024
Jessie Yuan, Janavi Gupta, Akhil Padmanabha, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, Zackory Erickson

[6] **Towards the Development of Wound Care Robots: An Observational Study Outlining Robotic Design Needs for Wound Care**
HRI 2024: Workshop on HRI for Aging in Place, 2024
Zulekha Karachiwalla, Ellen Lee, Andrew Dierkes, Zackory Erickson, Henny Admoni

[7] **Elastic Context: Encoding Elasticity for Data-driven Models of Textiles**
ICRA 2023: Embracing contacts. Making robots physically interact with our world, 2023
Alberta Longhini, Marco Moletta, Alfredo Reichlin, Michael C. Welle, Alexander Kravberg, Yufei Wang, David Held, Zackory Erickson, Danica Kragic

[8] **HAT: Head-Worn Assistive Teleoperation of Mobile Manipulators**
ICRA 2023: Emerging paradigms for assistive robotic manipulation: from research labs to the real world, 2023
Akhil Padmanabha*, Qin Wang*, Daphne Han, Jashkumar Diyora, Kriti Kacker, Hamza Khalid, Liang-Jung Chen, Carmel Majidi, Zackory Erickson

[9] **A Study of Causal Confusion in Preference-Based Reward Learning**
ICML 2022: Workshop on Spurious Correlations, Invariance and Stability, 2022
Jeremy Tien, Jerry Zhi-Yang He, Zackory Erickson, Anca Dragan, Daniel S. Brown